



C23-EE-304

23137

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DEEE – THIRD SEMESTER EXAMINATION

ELECTRONICS ENGINEERING

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :**
- (1) Answer **all** questions.
 - (2) Each question carries **three** marks.
 - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

- 1. Draw the V-I characteristics of P-N junction diode.
- 2. List any three applications of IGBT.
- 3. State the need of voltage regulator in power supplies.
- 4. Draw basic block diagram of regulated power supply.
- 5. Define the terms (a) Feedback factor and (b) Amplifier.
- 6. List different types of coupling methods in amplifiers.
- 7. State the conditions required for sustained oscillations.
- 8. List any three applications of oscillators.
- 9. State the concept of virtual ground.
- 10. List the characteristics of an ideal operational amplifier.

PART—B

10×5=50

- Instructions :**
- (1) Answer *any* **five** questions.
 - (2) Each question carries **ten** marks.
 - (3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

- 11. Explain the working of Zener diode and draw its V-I characteristics.
- 12. Explain and plot the input and output characteristics of transistor in CE configuration.
- 13. Explain the working and draw the circuit diagrams and waveforms of fullwave rectifier.
- 14. Explain the working of RC coupled amplifier with the circuit diagram and draw the frequency response curves.
- 15. (a) Explain the operation of transistor as an amplifier. 5
(b) State the need of an AF oscillator and RF oscillator. 5
- 16. Draw the circuit diagram and explain working of RC phase shift oscillator.
- 17. Explain the working of Op-Amp inverting amplifier.
- 18. Explain the operational amplifier as (a) Integrator and (b) Differentiator.

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